

A carbon market sensitive optimization model for integrated forward–reverse logistics

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ABSTRACT

Globalized supply chains, volatile energy and material prices, increased carbon regulations and competitive marketing pressure for environmental sustainability are driving supply chain decision makers to reduce carbon emissions. Enterprises face the necessity and the challenge of implementing strategies to reduce their supply chain environmental impact in order to remain competitive. One of the most important strategic issues in this context is the configuration of the logistics network. The decision concerning the design of an optimal network of the supply chain plays a vital role in determining the total carbon footprint across the supply chain and also the total cost. Therefore, the logistics network should be designed in a way that it could reduce both the cost and the carbon footprint across the supply chain. In this context, this research proposes a quantitative optimization model for integrated forward–reverse logistics with carbon-footprint considerations, by integrating the carbon emission into a quantitative operational decision-making model with regard to facility layout decisions. The proposed research incorporates carbon emission parameters with various decision variables and modifies traditional integrated forward/reverse logistics model into decision-making quantitative operational model, minimizing both the total cost and the carbon footprint. The proposed model investigates the extent to which carbon reduction requirements can be addressed under a particular set of parameters such as customer demand, rate of return of products etc., by selecting proper policy as an alternative to the costly investment in carbon-reducing technologies. To solve the quantitative model, this research implements a modified and efficient forest data structure to derive the optimal network configuration, minimizing both the cost and the total carbon footprint of the network. A comparative analysis shows the outperformance of the proposed approach over the conventional Genetic Algorithm (GA) for large problem sizes.

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1. Introduction

Reducing and mitigating carbon emission, the culprit of global warming and climate change, is an increasingly important concern for both industry and government (IPCC, 2007). In the UK, the government has targeted to reduce carbon emissions by 60% from 1990 levels by 2050 (Carbon Trust, 2006). The United Nations, the European Union, and many countries have enacted legislations or designed mechanisms such as carbon taxes, carbon offset, clean development, cap and trade, carbon caps and joint implementation to curb the total amount of carbon emissions. Firms worldwide, in response to such mechanisms and legislations or to concerns raised

by their own customers, are undertaking initiatives to reduce their carbon footprints.

However, most of the carbon emission reduction initiatives have focused largely on replacing energy inefficient equipment and facilities, redesigning products and packaging, finding less polluting sources of energy and implementing energy-saving programmes. While such efforts are valuable, many firms tend to ignore a potentially more significant source of emissions, which is driven by business practices, operational policies, interaction, and coordination in a long and complex supply chain, where the flow of products to consumers engages multiple firms (NSF Symposium report, 2010). For example, determining the frequency of supply and deliveries could be as important in reducing carbon emissions as the energy efficiency of the vehicles used to make these deliveries. Similarly, several decisions that a firm makes regarding its facility location, supplier and transportation mode selection, etc., can significantly affect its carbon footprint. In addition, in order to keep high service level, suppliers may require storing inventories in multiple locations that are close to

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their customers, increasing associated carbon footprint. It is therefore necessary to address the problem of carbon emissions reduction from supply chain and logistics perspective.

The Kyoto protocol introduced three major mechanisms for carbon emission reduction: namely carbon trading, clean development mechanism (CDM) and joint implementation (JI). The first mechanism serves as an economic incentive for companies to reduce carbon emissions. Governments set a specific limit of carbon emission for a particular company called carbon cap. If a company wishes to exceed the specified allowance, it will have to buy emission credits from companies that produce less GHGs or below their cap. The second and third mechanism allows countries to gain emission credits by financing emission reduction projects in developing countries and allows specified countries to carry out emission reduction projects in other countries to get emission credits. The carbon trading mechanism applies pressure on companies to reduce carbon emissions throughout their operations towards implementing more environmental friendly ways of conducting their business and to contribute to emission reduction projects worldwide. In addition, with the creation of the 'carbon market', competition amongst firms towards becoming 'greener' has become a major challenge.

Literature surveys conducted by [Hua et al. \(2011\)](#), [Abdallah et al. \(2010\)](#), [Sundarkani et al. \(2010\)](#), [Benjaafar et al. \(2013\)](#), and [Cachon \(2009\)](#) have identified a growing need for developing quantitative models and decision support systems for operations management, supply chain management, and logistics to address the issues associated with curbing carbon emissions. In addition, [Diabat and Simchi-Levi \(2009\)](#) highlight that integrating environmental issues into production, supply chains, and logistics is a complex process. However, with current financial incentives, branding benefits, governmental pressure, and the potential of creating a competitive edge, managers and practitioners have to move away from their current practices and identify ways to green their supply chains, a shift which is receiving increasing on-going research attention. In particular, there is a need for model-based research that extends and integrates current quantitative models which focus on minimizing cost and maximizing profit by including carbon-footprint criteria under different policy situations. These models could then be used to understand how carbon emissions parameters affect operational decisions. Such models can help policy makers to choose different policies related to reduction of carbon cap such as emission cap and trade, taxes on carbon emissions, etc. In addition, these models could also be used to understand how the specifics of these policies would affect the costs and carbon emissions of various firms.

Currently firms put more emphasis productivity and customer satisfaction, which leads firms to focus on their supply chain and integrated logistics ([Pishvae et al., 2009](#)). An efficient and robust logistics network can contribute towards a competitive advantage for firms and helps them to cope with increasing environmental and competitive pressures. Pressed by legislation and environmental consciousness while balancing economic benefits, several leading companies including General Motors, Kodak and Xerox have succeeded by focusing on their reverse logistics and recovery activities. Logistics network and its configuration are very important strategic issues in supply chain management and have a significant impact on the total performance of the supply chain. Decisions such as determining the numbers, locations and capacities of facilities and the quantity of flow between them are dependent on the logistic network design. Reverse logistics is evolving and requires setting up additional appropriate logistics infrastructure for arising flows of used and recovered products. Physical location, facilities and transportation links need to be chosen to transfer forward products from manufacturers to customers and to convey returned products from their former users to manufacturers for the purpose of recovery or safe disposal. Configuration of reverse logistics network has a strong influence

on the forward logistics network and vice-versa. Separating the design may lead to sub-optimality and therefore integration of design of forward and reverse logistics network is needed. Following the structure of integrated forward/reverse logistics model as presented in [Pishvae et al. \(2009\)](#), this research uses an integrated forward/reverse logistics network as shown in [Fig. 1](#).

The integrated logistics network shown above is a single product, multi-stage logistics network. It includes production, distribution, customer, collection, recovery and disposal centres. There are hybrid processing facilities such as distribution-cum-collection centre which is advantageous when compared to separate distribution or collection centre with regard to cost savings and carbon emission reduction. The main motivation behind this activity is sharing of material-handling equipment and infrastructure, prompting to use the above-mentioned integrated forward–reverse logistics network. New products are shipped from production-cum-recovery centre through distribution-cum-collection centre to customer zones while considering forward logistic. For reverse logistics, returned products are collected at distribution-cum-collection centre, then recoverable products are sent to production-cum-recovery centres and scrap products are sent to disposal centres. All these activities have significant impact on the carbon emission in the supply chain.

Aforementioned situation demands for the formulation of a comprehensive analytical model for systematically integrating forward and reverse logistics considering both the cost and the carbon emission criteria. Recognizing the inherent complexity of such integrated logistics network design problem, this paper extends the existing problem scenario, develops a mathematical model and proposes a solution methodology that can near-optimally minimize both the cost and the carbon emission considering various policies and criteria. We assume that some critical parameters such as demand and returned products are deterministic but customer demand, quantity and quality of returned product have high degree of uncertainty even in a short period of time. Depending on the degree of uncertainty, the level of carbon footprint can also vary. [Benjaafar et al. \(2013\)](#) showed that how important insights could be drawn by integrating carbon-emissions parameters into traditional and widely used lot-sizing models. This paper is a step forward contribution in this direction.

The decision regarding the design of an optimal network of the whole supply chain plays a major role in determining the total carbon footprint across the supply chain. This decision also involves a lot of investments. This research proposes an efficient quantitative optimization model for single period, single product integrated forward/reverse logistics network design, which takes into account both recovery and disposal activities. It further incorporates different carbon emission parameters into operational decision-making, which minimizes both total cost and total carbon footprint. This research further uses forest data structure algorithm proposed by [Delbem et al. \(2012\)](#) as a solution methodology to derive the optimal/suboptimal network configuration. Furthermore, we compare the results obtained using this solution methodology with the results obtained using Conventional Genetic Algorithm (CGA). A comparative study shows the outperformance of forest data structure algorithm for large sized problem with large number of operational nodes.

The remainder of the paper is organized as follows. [Section 2](#) systematically reviews the state of the art literature to identify research gaps and makes a case for the presented research. [Section 3](#) introduces and discusses the development of the proposed mathematical model of the problem under consideration with several defined policies and [Section 4](#) discusses the proposed solution methodology namely forest data structure algorithm and its application. [Section 5](#) presents data related to problems and computational experiments. [Section 6](#) presents numerical results and comparative study of various policies. A comparison of the proposed approach

with the CGA is also discussed in this section. Finally, Section 7 concludes this paper with a note for future research direction.

2. Literature review

The literature review of this research considers two main areas: (1) the advancement in research related to integrated forward and reverse logistics and (2) research related to carbon emission and environmental concern in the supply chain using quantitative models. It is evident from the literature review that there is extensive amount of research available in economics which deal with environmental concerns; however there are very little works related to quantitative models considering carbon emission. In one of the earlier work, Penkuhn et al. (1997) presented a nonlinear model for joint production scheme by integrating carbon taxes in the process industry and this model is implemented using a process simulation system. Simpson and Power (2005) found out the relationship between a supplier and firm's level of environmental management activity and the structure of the customer–supplier manufacturing relationship. Carbon Trust (2006) develops a procedure to track down the carbon emissions from various energies used across the supply chain. Mtalaa et al. (2009) reviews the present measurement and calculation methods and models available that compute CO₂ emissions from truck transportation. Cholette and Venkat (2009) evaluate the carbon emissions and energy associated with each transportation link in a wine supply chain. Sundarakani et al. (2010) provide an analytical model that considers carbon emissions from both stationary and non-stationary supply chain process. Kim et al. (2009) found out the relationship between freight transport costs and CO₂ emissions in given intermodal and truck freight network. A decision-support tool with multi-objective optimization capability was developed to show the ever changing trade-off and mode and route choice. Rosic et al. (2009) verify a single period dual model that incorporates the carbon emission cost and shows its effect on the offshore, onshore and total order quantity. Diabat and Simchi-Levi (2009) worked on both carbon sensitive supply chain network problem with green procurement and carbon capped supply chain. They assumed throughput capacity of the manufacturing site and storage capacity of the distribution centre as decision variables. Chaabane et al. (2012) presented a mixed integer linear programming (MILP) based framework that shows an efficient carbon management strategy which will help decision makers to achieve both sustainability objective and minimum cost simultaneously. Hua et al. (2011) show how carbon emission trading mechanisms can be used in inventory management. They derive optimal order quantity, and analytically examine the impacts of carbon trade, carbon price and carbon cap on order decisions, carbon emissions and total cost. However, this research did not consider the impact of various transportation modes and did not incorporate price decisions. Harris et al. (2011) worked on the relationship between total logistics costs and the CO₂ emissions from transportation and electricity. Kumar et al. (2012) argued using a case study that eliminating waste throughout the supply chain will make the supply chain more profitable. The outcome of this study highlights the importance for every company to apply green supply chain methods in order to stay or to become competitive.

Cabral et al. (2012) proposed a Lean, Agile, Resilient and Green (LARG) analytic network process (ANP) model for green supply chain. This model is validated using a case study of auto maker supply chain. Kumar et al. (2012) presented a case-study based simple model that companies can use to understand and improve supply chain sustainability practices. Ramudhin et al. (2008) developed a carbon market sensitive supply chain network problem. Zhang and Xu (2013) investigates the multi-item production planning problem with carbon cap and trade mechanism, where the firm can buy or sell the right to emit carbon on a trading market of carbon emission. They presented a profit-maximization model to

characterize the optimization problem and analysed the optimal policy of production and carbon trading decisions. Du et al. (2013), in their research, proved that the system-wide and the manufacturer's profits increase with the emission cap while the permit supplier's decreases. They suggested that there is room for manufacturer and permit supplier to coordinate the supply chain to get more profit in a certain condition. Benjaafar et al. (2013) considered three different carbon footprint policies and showed how carbon emission concerns can be integrated into operational decision-making. Most of these works incorporated various emission policies and emission criteria in their work however did not consider the integrated forward and reverse logistics aspect of supply chain.

Another series of research relating to forward and reverse logistics started with its integration in the context of supply chain. Sheu and Huang (2002) presented a multi-step reverse logistics system. Sheu et al. (2005) has developed a linear multi-objective model which optimizes the operations of both integrated logistics and reverse logistics in a given green supply chain. They used a linear multi-objective programming model for systematic optimization of integrated forward and reverse logistics in a given green supply chain. Vachon (2007) explores the possible relation between environmental collaboration and environmental monitoring in the supply chain and the form of environmental investment characterized by three categories namely pollution prevention, pollution control, and management systems. Sayed et al. (2010) proposed a stochastic model for forward–reverse logistic network design under risk. Few more relevant works related to recycling of products, reverse logistics including closed loop chains can be found in the work done by Giannikos (1998), Barros et al. (1998), Caruso et al. (1993), Listes and Dekker (2001), Jayaraman et al. (1999), Kroon and Vrijens (1995), Gottinger (1988) and Fleischmann et al. (2001).

Recently, Liu and Huang (2014) considered the production scheduling problem and incorporated carbon emission criteria in their objective model of minimizing the total weighted tardiness and carbon footprint simultaneously. They used a non-dominated sorting-based genetic algorithm II (NSGA-II) to solve this problem.

However, all the aforementioned research mainly focuses on the separate issues either relating to integrated forward and reverse logistics or incorporating carbon emission criteria/policies in supply chain. The literature tends to ignore the carbon emission and correspondingly various emission related regulatory policies in the context of integrated forward and reverse logistics. In fact, due to influence and contribution of the activities of reverse logistics and forward logistics on carbon emission, the integration of forward and reverse logistics needs to be considered with carbon emission and regulatory policy criteria. In addition, another inherent complexity associated with this problem lies in finding an efficient solution methodology. This research uses an innovative forest data structure method and applies it to solve the integrated forward and reverse logistics problem considering carbon emission. To the best of our knowledge, this research is another step forward in the application of forest data structure algorithm to solve a practical problem.

3. Problem formulation and mathematical model

This section discusses the problem formulation and presents our mathematical model to illustrate how carbon emissions considerations can be incorporated into operations management models. The model under consideration is a single period, single product integrated forward/reverse logistics network for a single firm with strict carbon caps. This choice of model is motivated mainly due to (1) the numerous uses of such model in practice, as they form the building blocks for many commercial supply chain planning decisions, (2) the ability of this model to accommodate many of the relevant concerns associated with carbon emissions parameters and related regulatory

policies. However, it is possible to develop models that are rooted in some other operations management models such as lot-sizing models, etc. To investigate carbon emissions concern, this research considers a single firm under strict carbon cap. Subsequently, it has been shown that by making small changes in the model, a firm can incorporate flexibility in carbon emissions using carbon trading mechanism, carbon cap-and-trade and carbon offsets.

A common problem faced by the firm is when to order and how much to order or when to produce and how much to produce under some specified decision variables over a specific planning horizon. This may lead to signification variation of carbon footprint across the whole supply chain. In absence of carbon emissions considerations, firms make ordering decisions to minimize the sum of its fixed (e.g., process and machine set-up cost) and variable costs (e.g., unit production or purchasing cost) to maximize its profit. However, in the presence of carbon emissions considerations, firms must look into the emissions associated with various decisions regarding ordering, production, inventory holding, etc. There may be variable carbon emissions associated with placing an order with a supplier (e.g., emissions due to transportation, emissions due to the handling or the production of each unit or emissions associated with storage of each unit held in inventory, etc.) or fixed carbon emissions (e.g., emission due to initiating production to process set-up, emissions due to set-up a plant or disposal centre, etc.).

Benjaafar et al. (2013) suggested that several regulatory policy settings can be considered in the model formulation to incorporate carbon emission concerns. These settings may include strict carbon caps (where firms are subject to mandatory caps on the amount of carbon they emit), carbon tax (firms are taxed for the amount of carbon they emit) and carbon cap and trade (firms are subject to carbon cap but rewarded or penalized for emitting less or more than tier cap). Our model is based on the model proposed by Benjaafar et al. (2013) and Pishvae et al. (2009). This research discusses the case of a single firm in formulating the model. Subsequently, we show that how flexibility can be added to the proposed model by incorporating carbon tax and cap-and-trade by modifying the objective functions and corresponding constraints. Next, we consider the situation presented in Fig. 1 to discuss the design of integrated logistics network by integrating the carbon emission criteria under various regulatory policy settings. Under this situation, the main issue to be addressed is to how to design an optimal network considering the location and number of hybrid distribution – collection, production recovery and disposal centre at the same time considering carbon emission and regulatory policies.

3.1. Model 1: A single firm with strict carbon caps

The following notations are used in the formulation of integrated logistics network.

3.1.1. Sets

- I: Set of potential production/recovery centre locations; $i \in I$
- J: Set of potential distribution-cum-collection centre locations; $j \in J$
- K: Set of fixed locations of customer zones; $k \in K$
- M: Set of potential disposal centre locations; $m \in M$

3.1.2. Parameters

- a_{jk} : Shipping cost per unit of products from distribution-cum-collection centre j to customer zone k
- b_{kj} : Shipping cost per unit of products from customer zone k to distribution-cum-collection centre j
- c_{ij} : Shipping cost per unit of products from production/recovery centre i to distribution-cum-collection centre j
- cw_i : Capacity of production per hour for production/recovery centre i
- cwr_i : Capacity of recovery per hour for production/recovery centre i
- cy_j : Capacity of handling products per hour in forward flow at distribution-cum-collection centre j
- cyr_j : Capacity of handling returned products per hour in reverse flow at distribution-cum-collection centre j
- cz_m : Capacity of handling scrapped products per hour at disposal centre m
- $CO_2^{\text{Forward}}_{\text{CAP}}$: Maximum amount (in tons) of carbon that can be emitted in forward logistic
- $CO_2^{\text{Reverse}}_{\text{CAP}}$: Maximum amount (in tons) of carbon that can be emitted in reverse logistic
- d'_k : Per hour demand of customer zone k
- d_{ij} : Distance (in miles) between production/recovery centre i and distribution-cum-collection centre j
- d_{jk} : Distance (in miles) between distribution-cum-collection centre j to customer zone k
- d_{jm} : Distance (in miles) between distribution-cum-collection centre j to disposal centre m
- e_{ji} : Shipping cost per unit of recoverable products from distribution-cum-collection centre j to production/recovery centre i
- f_i : Fixed cost of opening production/recovery centre i
- f'_i : Fixed carbon emission (in ton) from opening production/recovery centre i
- g_j : Fixed cost of opening distribution-cum-collection centre j
- g'_j : Fixed carbon emission (in ton) from opening distribution-cum-collection centre j
- h_m : Fixed cost of opening disposal centre m
- h'_m : Fixed carbon emission (in ton) from opening disposal centre m
- $N_{ij} = \lceil X_{ij}/\kappa_{pd} \rceil$: Number of trucks needed for shipping products from production/recovery centre i to distribution-cum-collection centre j
- $N_{jk} = \lceil U_{jk}/\kappa_{dc} \rceil$: Number of trucks needed for shipping products from distribution-cum-collection centre j to customer zone k

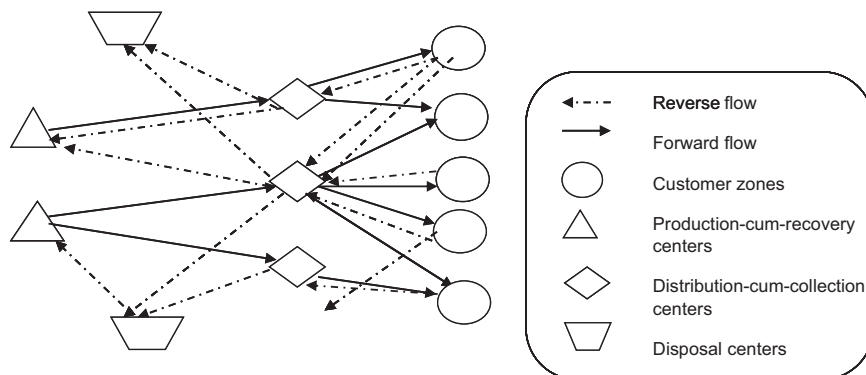


Fig. 1. Structure of integrated forward/reverse logistics network (Pishvae et al. (2009)).

$N_{kj} = \lceil Q_{kj} / \kappa_{cd} \rceil$: Number of trucks needed for shipping returned products from customer zone

k to distribution-cum-collection centre j

$N_{ji} = \lceil V_{ji} / \kappa_{dp} \rceil$: Number of trucks needed for shipping recoverable products from distribution-cum-collection centre j to production/recovery centre i

$N_{jm} = \lceil T_{jm} / \kappa_{dd} \rceil$: Number of truck needed for shipping scrapped products from distribution-cum-collection centre j to disposal centre m

P : Price of carbon per ton

p'_{jm} : Shipping cost per unit of scrapped products from distribution-cum-collection centre j to disposal centre m

r_k : Rate of return of used products from customer zone k

s : Average disposal fraction

α'_m : Penalty cost per unit of non-utilized capacity at disposal centre m

α_{ij} : Carbon emission (in ton per mile per unit product) during transportation from product/recovery centre i to distribution-cum-collection centre j

α_{jk} : Carbon emission (in ton per mile per unit product) during transportation from distribution-cum-collection centre j to customer zone k

α_{jm} : Carbon emission (in ton per mile per unit product) during transportation from distribution-cum-collection centre j to disposal centre m

β_j : Penalty cost per unit of non-utilized capacity at distribution-cum-collection centre j

β_{kj} : Carbon emission (in ton per mile per unit product) during transportation from customer zone k to distribution-cum-collection centre j

η_m : Disposal cost per unit of product at disposal centre m

ϕ_j : Processing cost per unit of product at distribution-cum-collection centre j

κ_{pd} : Truck capacity for production/recovery centre i to distribution-cum-collection centre j

κ_{dc} : Truck capacity for distribution-cum-collection centre j to customer zone k

κ_{cd} : Truck capacity for customer zone k to distribution-cum-collection centre j

κ_{dp} : Truck capacity for distribution-cum-collection centre j to production/recovery centre i

κ_{dd} : Truck capacity for distribution-cum-collection j to disposal centre m

$\gamma^{pro/rec}$: Carbon emission (in ton per unit product) due to manufacturing/recovery of a particular product type

γ^{dis} : Carbon emission (in ton per unit product) due to disposal of a particular product type

ρ_i : Manufacturing/recovery cost per unit of product at product at production/recovery centre i

τ_i : Penalty cost per unit of non-utilized capacity at production/recovery centre i

3.1.3. Decision variables

Q_{kj} : Quantity of returned products shipped from customer zone k to distribution-cum-collection centre j

T_{jm} : Quantity of scrapped products shipped from distribution-cum-collection centre j to disposal centre m

U_{jk} : Quantity of products shipped from distribution-cum-collection centre j to customer zone k

V_{ji} : Quantity of recoverable products shipped from distribution-cum-collection centre j to production-cum-recovery centre i

$W_i = 1$; if production-cum-recovery centre is opened at location i
0; otherwise

$Y_j = 1$; if distribution-cum-collection centre is opened at location j
0; otherwise

0; otherwise

$Z_m = 1$; if disposal centre is opened at location i

0; otherwise

X_{ij} : Quantity of products shipped from product in-cum-recovery centre to distribution-cum-collection centre j

3.1.4. Objective function

Using the above-mentioned notations, the carbon-footprint optimization model for integrated forward–reverse logistics can be formulated as follows:

Minimize

$$\begin{aligned} & \sum_{i \in I} (f_i + P \times f'_i) \times W_i + \sum_{j \in J} (g_j + P \times g'_j) \times Y_j + \sum_{m \in M} (h_m + P \times h'_m) \times Z_m \\ & + \sum_{i \in I} \sum_{j \in J} [X_{ij} \times (\rho_i + c_{ij}) + P \times \gamma^{pro/rec}] + P \times d_{ij} \times \alpha_{ij} \times N_{ij} \\ & + \sum_{j \in J} \sum_{k \in K} [U_{jk} \times ((\phi_j + a_{jk}) + P \times d_{jk} \times \alpha_{jk} \times N_{jk})] + \sum_{k \in K} \sum_{j \in J} [Q_{kj} \\ & \times ((\phi_j + b_{kj}) + P \times d_{kj} \times \beta_{kj} \times N_{kj})] + \sum_{j \in J} \sum_{i \in I} [V_{ji} \times (\rho_i + e_{ji} \\ & + P \times \gamma^{pro/rec}) + P \times d_{ij} \times \alpha_{ij} \times N_{ij}] + \sum_{j \in J} \sum_{m \in M} [T_{jm} \times (\eta_m + p'_{jm} + P \times \gamma^{dis}) \\ & + P \times d_{jm} \times \alpha_{jm} \times N_{jm}] + \sum_{m \in M} \alpha'_m [(Z_m - cZ_m) + \sum_{j \in J} T_{jm}] + \sum_{i \in I} \tau_i \\ & \times [(W_i \times cw_i - \sum_{j \in J} X_{ij})] + (W_i \times cwr_i - \sum_{j \in J} V_{ji}) + \sum_{j \in J} \beta'_j \times [(Y_j \times cy_j \\ & - \sum_{k \in K} U_{jk}) + (Y_j \times cyr_i - \sum_{k \in K} Q_{kj})] \end{aligned} \quad (1)$$

In this objective function (Eq. (1)), first, second and third terms minimize fixed cost and fixed carbon emissions for opening a production-cum-recovery plant, distribution-cum-collection centre and disposal centre respectively. Fourth and fifth terms consider the variable cost and variable emissions associated with production, transportation, processing for an unit product in forward logistics. Sixth, seventh and eighth terms represent variable cost and variable emissions due to collection, recovery, transportation of a unit product for reverse logistics. Rest of the terms are used to assign penalty costs for non-utilized capacities at disposal centre, production-cum-recovery centre and distribution-cum-collection centre respectively. The following constraints need to be satisfied simultaneously for this optimization model.

3.1.5. Subject to the following constraints

$$\begin{aligned} & \sum_{i \in I} f'_i \times W_i + \sum_{j \in J} g'_j \times Y_j + \sum_{m \in M} h'_m \times Z_m + \sum_{i \in I} \sum_{j \in J} [(X_{ij} \times \gamma^{pro/rec}) \\ & + (d_{ij} \times \alpha_{ij} \times N_{ij})] + \sum_{j \in J} \sum_{k \in K} [d_{jk} \times \alpha_{jk} \times N_{jk}] \leq CO2^{FORWARD}CAP \end{aligned} \quad (2)$$

$$\begin{aligned} & \sum_{k \in K} \sum_{j \in J} [d_{kj} \times \beta_{kj} \times N_{kj}] + \sum_{j \in J} \sum_{i \in I} [d_{ij} \times \alpha_{ij} \\ & \times N_{ji} + \sum_{j \in J} \sum_{m \in M} [(T_{jm} \times \gamma^{dis}) + (d_{jm} \times \alpha_{jm} \\ & \times N_{jm})] \leq CO2^{REVERSE}CAP \end{aligned} \quad (3)$$

$$\sum_{j \in J} U_{jk} \geq d'_k \quad \forall k \in K \quad (4)$$

$$\sum_{j \in J} Q_{kj} \geq r_k \times d'_k \quad \forall k \in K \quad (5)$$

$$\sum_{i \in I} X_{ij} - \sum_{k \in K} U_{jk} = 0 \quad \forall j \in J \quad (6)$$

$$\sum_{i \in I} V_{ji} - (1-s) \sum_{k \in K} Q_{kj} = 0 \quad \forall j \in J \quad (7)$$

$$\sum_{m \in M} T_{jm} - s \sum_{k \in K} Q_{kj} = 0 \quad \forall j \in J \quad (8)$$

$$\sum_{j \in J} V_{ji} - \sum_{j \in J} X_{ij} \leq 0 \quad \forall i \in I \quad (9)$$

$$\sum_{j \in J} X_{ij} \leq cw_i \times W_i \quad \forall i \in I \quad (10)$$

$$\sum_{i \in I} X_{ij} \leq cy_j \times Y_j \quad \forall j \in J \quad (11)$$

$$\sum_{k \in K} Q_{kj} \leq cyr_j \times Y_j \quad \forall j \in J \quad (12)$$

$$\sum_{j \in J} V_{ji} \leq cwr_i \times W_i \quad \forall i \in I \quad (13)$$

$$\sum_{j \in J} T_{jm} \leq cz_m \times Z_m \quad \forall m \in M \quad (14)$$

$$W_i, Y_j, Z_m \in \{0, 1\} \quad \forall i \in I, j \in J, m \in M \quad (15)$$

$$X_{ij}, U_{jk}, Q_{kj}, V_{ji}, T_{jm}, N_{ij}, N_{jk}, N_{kj}, N_{ji}, N_{jm}, CO_{2CAP}^{Forward}, CO_{2CAP}^{Reverse} \geq 0 \quad \forall i \in I, j \in J, k \in K, m \in M \quad (16)$$

Eqs. (2) and (3) restrict the total carbon emissions less than or equal to the carbon cap specified for forward and reverse logistic networks respectively. Eqs. (4) and (5) ensure that demand of all customer zones are satisfied and returned products from all customer are collected. Eqs. (6)–(9) confirm the flow balance at production-cum-recovery centre and distribution-cum-collection centre in forward and reverse flows. Eqs. (10)–(14) represent capacity constraints on facilities. Finally Eqs. (15) and (16) enforce binary and non-negativity restrictions on corresponding decision variables.

Here it is important to mention that the proposed model does not deal with the stochastic environment, but it is quite obvious that in real life situation one may have to deal with uncertain customer demands. In addition, last decade saw a significant fluctuation in fuel price, which effects transportation cost. Therefore, depending on the order size, type of vehicle, type of fuel used and traffic condition for a particular scenario (e.g., during peak hours shorter route may be congested and to avoid that and to ensure quick delivery one may need to drive through longer route to reach the same destination, leading to increase in both the transportation cost and the carbon emission), transportation cost as well as carbon emissions will significantly vary. Similarly, depending on the process, machines or technologies involved (e.g., for large number of products we use some machine set-up and for relatively smaller number we use some other set-up), manufacturing/recovery cost, processing cost, disposal cost per unit of products may vary under a particular scenario. Our model could be extended to incorporate these uncertainties. In this model, by using binary variables one can decide which plant-cum-recovery centre or distribution-cum-collection centre or disposal centre to be opened or to be closed based on the data available for a particular scenario in order to minimize both total cost and total carbon footprint across the supply chain.

In the abovementioned formulation, we assume that there is no limitation or constraint on order sizes and there are no supply lead times. But we can consider both of them by slightly modifying our model. It is also possible to incorporate appropriate constraints which enable us to make sure that demand in each period must be fulfilled. Here, we have considered carbon emission cap over the

entire planning horizon but it is possible to model the system such that the cap is over smaller subsets of periods or is associated with each small period. The emission cap could be also associated with each unit produced or ordered, for example, firms may wish to market those products whose carbon footprint per unit do not exceed a certain threshold. While defining carbon emission parameters, we assume that emission is linearly increased with the decision variables however one can show increase in emission by some non-linear functions by incorporating slight changes in the model (Here it can be noted that a non-linear function can be approximated by a piece-wise linear function.). We assume that the carbon emission parameters are readily available (Which is not the case for many practical scenarios due to several public policies and lack of statistical tools. In addition, quantitative measurement of carbon emission in itself is a huge task for any firm.). More recently significant effort is made by several firms to document the carbon footprints of their activities, as this is required in documenting their compliance with regulatory policies or in communicating the carbon footprint of their products to consumers.

Finally, it might be tempting to view carbon cap as being similar to capacity limits on production or ordering, which are most common in many operations model but they differ in several point of views: First, carbon cap may cover multiple periods or even the entire planning horizon whereas production capacity typically applies to each period. Second, carbon capacity is consumed not only by production, order processing and transportation but it can also be consumed by simply holding inventory even if there is no production. Therefore, it also means that operating changes in one area (say inventory) may induce changes in demand or in other areas such as production or transportation due to joint carbon capacity limit. Third, carbon capacity consumed in one period may affect the available carbon capacity in future periods (in case of a carbon cap over the entire planning horizon) whereas production usually regenerates in each period. Next, we discuss another integrated forward and reverse logistics network model for a single firm, which is taxed for the amount of carbon it can emit (i.e., carbon tax).

3.2. Model 2: A single firm with carbon tax

There can be an alternative to strict caps on emission, which is to penalize emission using carbon tax instead of restricting emissions within a certain level. There may be variety of ways to incorporate carbon tax into the model among which the simplest form is to consider the financial penalty linearly with the number of carbon units emitted. To illustrate, how the carbon tax can be incorporated into the model proposed in Section 3.1, we follow the model proposed by Benjaafar et al. (2013) and Pishvae et al. (2009) and modify it in the following way.

Let, p' denote the amount of tax paid on each unit of carbon emission, therefore the objective function can be modified as

3.2.1. Objective function

Minimize

$$\begin{aligned} & \sum_{i \in I} f_i \times W_i + \sum_{j \in J} g_j \times Y_j + \sum_{m \in M} h_m \times Z_m + \sum_{i \in I} \sum_{j \in J} X_{ij} \times (\rho_i + c_{ij}) \\ & + \sum_{j \in J} \sum_{k \in K} U_{jk} \times (\varphi_j + a_{jk}) + \sum_{k \in K} \sum_{j \in J} Q_{kj} \times (\varphi_j + b_{kj}) + \sum_{j \in J} \sum_{i \in I} V_{ji} \\ & \times (\rho_i + e_{ji}) + \sum_{j \in J} \sum_{m \in M} T_{jm} \times (\eta_m + p'_{jm}) + \sum_{m \in M} \alpha'_m [(z_m - cz_m) \\ & + \sum_{j \in J} T_{jm}] + \sum_{i \in I} \tau_i \times [(W_i \times cw_i - \sum_{j \in J} X_{ij}) + (W_i \times cwr_i - \sum_{j \in J} V_{ji})] \\ & + \sum_{j \in J} \beta'_j \times [(Y_j \times cy_j - \sum_{k \in K} U_{jk}) + (Y_j \times cyr_j - \sum_{k \in K} Q_{kj})] + P' \times [\sum_{i \in I} f'_i \\ & \times W_i + \sum_{j \in J} g'_j \times Y_j + \sum_{m \in M} h'_m \times Z_m + \sum_{i \in I} \sum_{j \in J} (X_{ij} \times \gamma^{pro/rec}) + d_{ij} \end{aligned}$$

$$\begin{aligned}
& \times \alpha_{ij} \times N_{ij}] + \sum_{j \in J} \sum_{k \in K} (d_{jk} \times \alpha_{jk} \times N_{jk}) + \sum_{k \in K} \sum_{j \in J} d_{kj} \times \beta_{kj} \times N_{jk} \\
& + \sum_{j \in J} \sum_{i \in I} [(V_{ji} \times \gamma^{pro/rec}) + d_{ij} \times \alpha_{ij} \times N_{ji}] \\
& + \sum_{j \in J} \sum_{m \in M} [(T_{jm} \times \gamma^{dis}) + d_{jm} \times \alpha_{jm} \times N_{jm}] \quad (17)
\end{aligned}$$

Subject to the constraints mentioned in Eqs. (4)–(16) [Note that here we do not include Eqs. (2) and (3) which are exclusively for strict carbon cap model.]. Several variations of this model are possible such as tax penalties which are nonlinear in emission quantities, tax schedule in which unit penalty changes in discrete steps as a function of emission quantity including the case where taxes are not levied if emissions do not fall below a certain threshold. If some of the variations of carbon tax lead to nonlinear optimization problem then it is possible to linearize the model by approximating the tax penalties with piecewise linear function. Next we extend our proposed model to incorporate carbon cap and trade (in which firms are subject to carbon cap but rewarded or penalized for emitting less or more than tier cap).

3.3. Model 3: A single firm with carbon cap-and-trade:

Apart from either restricting carbon emissions by imposing carbon cap or applying carbon tax, there may be an alternative where firms are allowed to emit more than their prescribed caps but are penalized for doing so with penalties increasing in the amount of emissions that exceed the cap. Firms are also rewarded for emitting less than their caps by receiving payments increasing in the difference between their caps and their actual emissions. This system can be implemented directly by a regulatory body or indirectly with the help of a trading market for carbon emissions, in which firms can buy and sell the right to emit.

If we assume that the carbon price is relatively stable over the firm's entire planning horizon and is exogenous to the decisions made by the firm, then based on the model proposed by Benjaafar et al. (2013) and Pishvaei et al. (2009), we can reformulate the given problem as follows:

3.3.1. Objective function

Minimize

$$\begin{aligned}
& \sum_{i \in I} f_i \times W_i + \sum_{j \in J} g_j \times Y_j + \sum_{m \in M} h_m \times Z_m + \sum_{i \in I} \sum_{j \in J} X_{ij} \times (\rho_i + C_{ij}) \\
& + \sum_{j \in J} \sum_{k \in K} U_{jk} \times (\varphi_j + a_{jk}) + \sum_{k \in K} \sum_{j \in J} Q_{kj} \times (\varphi_j + b_{kj}) + \sum_{j \in J} \sum_{i \in I} V_{ji} \\
& \times (\rho_i + e_{ji}) + \sum_{j \in J} \sum_{m \in M} T_{jm} \times (\eta_m + p_{jm}) + \sum_{m \in M} \alpha_m' [(Z_m - CZ_m) + \sum_{j \in J} T_{jm}] \\
& + \sum_{i \in I} \tau_i \times [(W_i \times cw_i - \sum_{j \in J} X_{ij}) + (W_i \times cwr_i - \sum_{j \in J} V_{ji})] + \sum_{j \in J} \beta_j' \\
& \times [(Y_j \times cy_j - \sum_{k \in K} U_{jk}) + (Y_j \times cyr_i - \sum_{k \in K} Q_{kj})] \\
& + P' \times (b^+ - s^-) \quad (18)
\end{aligned}$$

3.3.2. Subject to the following constraints

$$\begin{aligned}
& \sum_{i \in I} f_i' \times W_i + \sum_{j \in J} g_j' \times Y_j + \sum_{m \in M} h_m' \times Z_m + \sum_{i \in I} \sum_{j \in J} [(X_{ij} \times \gamma^{pro/rec}) \\
& + d_{ij} \times \alpha_{ij} \times N_{ij}] + \sum_{j \in J} \sum_{k \in K} (d_{jk} \times \alpha_{jk} \times N_{jk}) + \sum_{k \in K} \sum_{j \in J} d_{kj} \times \beta_{kj} \\
& \times N_{jk} + \sum_{j \in J} \sum_{i \in I} [(V_{ji} \times \gamma^{pro/rec}) + d_{ij} \times \alpha_{ij} \times N_{ji}] + \sum_{j \in J} \sum_{m \in M} [(T_{jm} \times \gamma^{dis}) \\
& + d_{jm} \times \alpha_{jm} \times N_{jm}] + s^- \leq (CO2^{FORWARD} CAP + CO2^{REVERSE} CAP) + b^+ \quad (19)
\end{aligned}$$

$$b^+, s^- \geq 0 \quad (20)$$

along with other constraints listed in Eqs. (4)–(16).

Here, b^+ and s^- represent the amount of carbon credit the firm buys and sells respectively. b^+ i.e., the carbon credits bought serve to relax the effective cap imposed on emissions although it adds up to the cost component of the objective function while s^- i.e., the carbon credits sold acts a new source of revenue.

In case of carbon trading, (1) carbon price is affected by market dynamics i.e., it will be subject to volatility and (2) the total amount of carbon that can be bought and sold is limited by the sum of the carbon caps imposed on the firms participating in the carbon trading system. However in the above formulation, it has been assumed that the market price for carbon is fixed. The volatility can be assumed away if it is of short-time nature and the long-term trend can be reliably forecast. Also, with the help of different financial options like call and put options, which are commonly used in the procurements of the commodity or which ensures that the firm can buy or sell commodity at a specified price, a firm can reduce the effect of price fluctuation. However, sometimes we need to construct a model which explicitly accounts for price volatility or else price would be sensitive to the supply and demand of carbons which is itself dependant on the imposed carbon caps. For this purpose, one can use the carbon offset model as proposed by Benjaafar et al. (2013) in their research. In addition, if there is a limit on how much carbon credit a firm can buy or sell in a particular time period then we can impose an upper bound on b^+ and s^- and modify the model accordingly or if there is a need to differentiate between the price of buying and selling of carbon we can modify the objective function by incorporating prices P^+ and P^- associated with b^+ and s^- respectively. We can also introduce period-specific emission trading variables to show that the model can be extended to settings where the emission price is period dependant or where the cap is imposed over intervals shorter than the entire planning horizon.

Next section discusses the forest data structure algorithm and its application to solve the proposed model in Section 3.

4. Forest data structure based solution methodology

In this research, we use forest data structure algorithm as proposed by Delbem et al. (2012) to solve our model. The main motivation to use forest data structure algorithm lies in its properties, which promise to have several benefits to solve a complex problem like the proposed one. Node depth degree representation (NDDR), a new dynamic forest data structure that constructs and manipulates a set of spanning forests of sparse graphs has an average running time of $O(\sqrt{n})$. A detailed study of forest data structure algorithm is presented in Delbem et al. (2012) and therefore for the sake of completeness we have only presented a brief overview. Readers are referred to Delbem et al. (2012) for more information.

The NDDR of a tree T consists of an ordered list (or array) of triples which contains tree nodes, tree depth and their degree in T . Given a tree, depth first search algorithm can be used to obtain a NDDR by adding a node, its depth and degree to the end of NDDR when search finds an unvisited node. The next step involves forest generation procedure to generate new spanning forest. The main reason to use forest generation process is to transfer a subtree from some tree T_{from} to another tree T_{to} in the forest. This type of tree modification has extensively been used in the areas such as genetic programming. Delbem et al. (2012) has used two different operations for generating new forest i.e., solution. Between two proposed operations in that paper, this research uses operation 1 only to generate new spanning forests. Operation 1 produces simple and small changes in the forest and has the following properties:

1. It maintains the consecutiveness of the node in node-depth-degree representation (NDDR).

2. It maintains the precedence of the root of a sub-tree as regards to its descendants.
3. Does not introduce cycle in the trees.
4. Maintains the number of trees in the forest.

The basic working principle of operation 1, as mentioned in the paper by Delbem et al. (2012), is shown in the following figures. This operation has been used in the present research to generate new forest (solution).

In Fig. 2(a) we have shown two trees and their node-depth degree representations (NDDRs). T_{from} corresponds to the tree from which we will prune an arc (from point p) and then the pruned arc is stored temporarily in T_{temp} . Finally, we will attach T_{temp} to T_{to} at point a. The output is a new forest F_h that contains T_{to} and T_{from} as shown in 2(c). The NDDRs of T_{to} and T_{from} are also shown in 2(c).

However, this operation does not consider the case of overlapping of arcs. However, in the proposed model due to the nature of the problem few nodes are hybrid nodes and they deal with both forward and reverse logistics. Therefore, some situations may arise where there is an overlap in the tree. To solve this problem, the entire network is divided into six levels as shown below and then performs operation 1. In this manner, the working principle of the algorithm remains the same but it ensures that there is no overlapping in the tree.

This research has formulated the problem accordingly and then each of the facilities is represented using Node-Depth-Degree Representation. Next, an initial degree unconstrained minimum spanning tree network based on capacity constraints has been formed. Allocation of different 'units of products' along different paths of the tree based on appropriate nodes of lower or higher level as applicable has been done. Next, we have reformed (using option 1) the tree without violating the capacity constraints. The representation of each facility using NDDR ensures the smoother and quicker calculation during new spanning tree generation and that is why this representation is used to encode or represent our facilities (Fig. 3).

The basic steps involved in the algorithm are as follows:

- Step 1:** Initialize the nodes at each level (capacity allocation)
- Step 2:** Form an initial degree unconstrained minimum spanning tree network based on capacity constraints.
- Step 3:** Allocate 'units of products' along different paths of the tree based on appropriate nodes of lower or higher level as applicable.
- Step 4:** Reform (using option 1) the tree without violating the capacity constraints.
- Step 5:** Repeat steps 3 and 4 until you reach to termination criteria i.e., no improvement of solution.

Next section discusses the computational experiments using the abovementioned algorithm.

5. Computational experiments

In this section, we summarize our computational experiments with the algorithms outlined in the previous section. All the instances were solved in 1000 iterations on a HP i7 M640 2.80 GHz with 4 GB RAM PC running the Windows 7 operating system.

5.1. Experimental data

This research has partly used the data as described in Pishvaei et al. (2009) to design the computational experiment for the proposed model. Table 1 shows the distances between distribution-cum-collection centre and production centre.

Table 2 shows the distances between various customer zones and distribution-cum-collection centres.

Table 3 presents various shipping costs associated with the products that are to be transported among various facilities throughout the supply chain.

Table 4 presents the demand of products across various customer zones.

Table 5 shows the quantity rate of return for products from various customer zones during reverse logistic.

Table 6 presents the fixed costs to open different facilities and capacity of each of the facilities.

Table 7 shows the amount of carbon that gets emitted during set up of each facility as well as amount of carbon emissions that are associated with transporting and processing of products.

Table 8 presents the penalty costs associated with different facilities as well as manufacturing/recovery and processing costs associated with products.

Table 9 presents data related to different policies such as carbon emission limits for both forward and reverse logistics under strict carbon policy, carbon price, amount of carbon that can be bought or sold under carbon cap and trade policies.

Truck capacities for various facilities are shown in Table 10. Generally, a large number of products are transported at a time from production centre to distribution-cum-collection centre. Therefore, larger capacity trucks are used in this case. Relatively less numbers of products are transported at a time to customer zones, so smaller capacity trucks are used in this case. For reverse logistics, generally smaller numbers of parts are transported in between the facilities, so trucks having very small capacities are used in this case.

Average Disposal Rate is taken as 0.2 for all the cases.

Next section discusses the results obtained using the abovementioned data sets and the application of NDDR method.

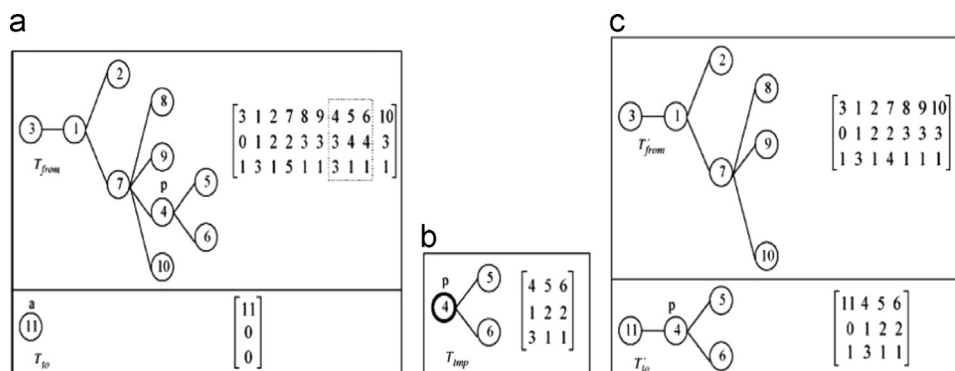


Fig. 2. Example of the application of Operation 1 [adapted from Delbem et al. (2012)].

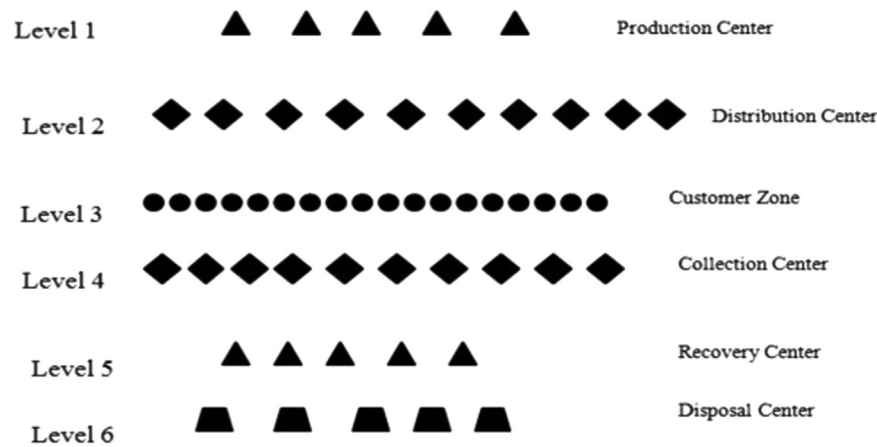


Fig. 3. Splitting of the network into six different levels.

Table 1

Distance (in miles) Matrix for production-cum-recovery centre (P) to distribution-cum-collection (DC) centre.

P	DC									
	1	2	3	4	5	6	7	8	9	10
1	5	4	6	9	4	5	11	15	12	10
2	4	12	10.4	3.4	2.8	8.4	7.5	6.5	4.7	7.3
3	5.5	5.6	3.6	5.9	6.4	5.6	7.3	6.3	6.7	5.6
4	6	4.5	7.8	6.1	5.9	4.5	5.7	7.8	8.9	4.8
5	3	7.9	3.4	7.2	6.8	4.8	7.2	9.3	5	7.5

Table 2

Distance (in miles) Matrix for distribution-cum-collection (DC) centre to customer zone (k).

K	DC																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	5	4	7	2	4	12	7	10	5	2	4	7	9	3	12	5	3	7	3	10
2	5	4	9	8	4	8	7	9	5	4	4	7	9	5	8	5	5	7	5	8
3	2	4	7	9	3	2	6	11	5	4	9	6	5	10	10	5	6	9	5	8
4	2	6	8	8	5	12	8	10	2	5	4	8	8	3	9	2	5	6	5	10
5	5	4	8	8	3	9	9	12	5	5	4	9	6	5	10	5	3	8	5	12
6	3	5	6	7	3	10	9	11	3	4	7	9	8	3	12	3	3	5	4	4
7	2	6	7	7	5	8	9	11	2	5	4	9	6	5	10	2	2	5	4	9
8	4	4	7	9	5	8	7	11	4	4	4	7	9	5	8	4	7	2	5	4
9	5	4	9	6	5	10	8	5	5	4	8	8	3	9	5	3	5	5	4	4
10	4	7	9	8	3	2	9	12	4	5	4	9	6	5	10	4	2	3	4	7

Table 3

Shipping costs (\$ per unit of product).

Production/recovery centre to distribution-cum-collection centre (c_{ij})	23
Distribution-cum-collection centre to customer zone (a_{jk})	15
Customer zone to distribution-cum-collection centre (b_{kj})	11
Distribution-cum-collection centre to production/recovery centre (e_{ji})	17
Distribution-cum-collection centre to disposal centre (p'_{jm})	7

Table 4

Demands (no of units/hour) of customer zones.

d_1	d_2	d_3	d_4	d_5	d_6	d_7	d_8	d_9	d_{10}
4900	4200	4180	4190	4550	4470	4600	4620	4650	4210
d_{11}	d_{12}	d_{13}	d_{14}	d_{15}	d_{16}	d_{17}	d_{18}	d_{19}	d_{20}
4570	4690	4190	4800	4810	4990	4250	4240	5000	4120

* d_k refers to demand of customer zone k .

Table 5

Quantity rate of return (Q_k) from customer zones.

Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8	Q_9	Q_{10}
0.7	0.72	0.69	0.71	0.73	0.67	0.75	0.72	0.7	0.73
Q_{11}	Q_{12}	Q_{13}	Q_{14}	Q_{15}	Q_{16}	Q_{17}	Q_{18}	Q_{19}	Q_{20}
0.72	0.69	0.71	0.73	0.67	0.75	0.71	0.74	0.68	0.72

6. Results and discussion

Computational result (Table 11) shows that for the given set of data, if the firm chooses policy 1 i.e., impose strict carbon cap on its carbon emission, it incurs minimal amount of carbon emission (as shown in Fig. 4) compared to the two other policies. However, the total cost (Fig. 5) incurred in this case is more than the cost incurred in two other policies. For the given set of data, this result could be justified because, for policy 1, the firm cannot emit more carbon than its specified limit (carbon cap) and for this purpose the firm may need to adopt expensive routes, vehicles and operate through other facilities which have higher operating cost but emit lower carbon.

In our problem formulation, we considered several decision variables, which include binary decision variables for determining whether one should open or close a facility. After solving the problem, one array of solution against each binary decision variable is obtained, satisfying all the constraints. The array contains a combination of '0' and '1'. '0' corresponds to closing of a particular facility and '1' corresponds to opening of a particular facility as shown in Table 11.

It can be seen from Fig. 4 that the total carbon emission can be minimized for a particular situation or set of parameters by choosing proper policies. Therefore, the decision makers of the firms can decide what are the policies (under the specified range of certain parameters) that need to be chosen well in advance to minimize both cost as well as carbon footprint.

On the other hand, by using policies 2 and 3, the firm has the flexibility to buy and sell carbon in the 'carbon market'. Thus, it incurs lower cost but higher carbon emission compared to policy 1. It is up to the decision makers of the firm to decide which policy to follow depending on whether their priority is to minimize the total cost or to minimize the total carbon emission. For example, if one wants to minimize the total cost, policy 3 will be the best option in this case. Similarly, if a decision maker wants to minimize the total carbon emission, policy 1 will be the best option to choose. However, all these decisions are based on the particular data set as presented in the earlier section.

Table 6
Fixed costs, capacity of different facilities.

Serial no.	f_i (\$ $\times 10^5$)	cw_i (units $\times 10^3$ /h)	cwr_i (units/h)	g_j (\$ $\times 10^4$)	cy_j (units/h)	cyr_j (units/h)	h_m (\$ $\times 10^4$)	cz_m (units/h)
1	60	12	75	60	1000	20	56	50
2	40	10	50	40	900	30	55	43
3	65	13	43	65	1500	25	57	70
4	75	23	70	75	1200	35	52	65
5	62	18	45	62	800	40	53	62
6	–	–	–	52	1100	15	–	–
7	–	–	–	72	700	23	–	–
8	–	–	–	34	950	27	–	–
9	–	–	–	53	850	31	–	–
10	–	–	–	45	750	28	–	–

Table 7
Data related to fixed and variable carbon emission.

Serial no.	f_i (ton)	g_j (ton)	h_m (ton)	α_{ij} (ton per mile per unit)	α_{jk} (ton per mile per unit)	α_{jm} (ton per mile per unit)	β_{kj} (ton per mile per unit)	$\gamma^{\text{pro/rec}}$ (ton per unit product)	γ^{dis} (ton per unit product)
1	50	45	56	0.025	0.025	0.015	0.015	0.005	0.008
2	60	35	55	–	–	–	–	–	–
3	45	42	57	–	–	–	–	–	–
4	55	41	53	–	–	–	–	–	–
5	35	37	54	–	–	–	–	–	–
6	–	39	–	–	–	–	–	–	–
7	–	38	–	–	–	–	–	–	–
8	–	40	–	–	–	–	–	–	–
9	–	32	–	–	–	–	–	–	–
10	–	33	–	–	–	–	–	–	–

Table 8
Data related to penalty costs at different facilities, manufacturing/recovery and processing costs.

Prob. size ($F^*J^*K^*M$)	α'_m (\$ per unit)	β'_j (\$ per unit)	η_m (\$ per unit)	τ_l (\$ per unit)	ϕ_j (\$ per unit)	ρ_l (\$ per unit)
5*10*20*5	5	8	5	3	15	18

Table 9
Data related to carbon cap and carbon price.

$CO_{ZAP}^{\text{FORWARD}}$	600 t
$CO_{ZAP}^{\text{REVERSE}}$	100 t
Carbon price (\$/ton)	50
Carbon credit bought (b^+)	75 t
Carbon credit sold (s^-)	30 t

Table 10
Data related to truck capacity.

Truck going from	Truck capacity
Production/recovery centre to distribution-cum-collection centre	3000 units
Distribution-cum-collection centre to customer zones	500 units
Customer zones to distribution-cum-collection centre	50 units
Distribution-cum-collection centre to production/recovery centre	50 units
Distribution-cum-collection centre to disposal centre	50 units

If one changes some of the data, the optimal network configuration may be completely different and thus the result may be different, so a decision maker has to decide based on the varied set

of data. For example when the model was tested by changing the few parameters to $CO_{ZAP}^{\text{FORWARD}} = 700$ t, $CO_{ZAP}^{\text{REVERSE}} = 200$ t, carbon credit bought (b^+) = 100 t, Carbon credit sold (s^-) = 35 t, policy 1 showed the lowest cost and policy 2 gave the lowest carbon emission while comparing with other policies as shown in Figs. 6 and 7.

Fig. 8 shows the impact of varying the carbon cap on total cost and total carbon emissions. As mentioned earlier, it should be noted that while we vary the carbon cap, the optimal network configuration may be changed in some of the cases, so the plot presented is not for a particular optimal network configuration. As expected, reducing the carbon emission cap increases total cost and reduces total carbon emissions. But an interesting phenomenon is observed in this plot, which shows that if we reduce the carbon cap from 1904 to 1525, one gets a carbon emission reduction of 31.19% but with a small increase in total cost by 1.79%. Therefore, it is possible to significantly reduce carbon emissions without significantly increasing the total cost.

In addition, in order to justify the use of NDDR based forest data structure this research compared the performance of proposed algorithm with the well-known and used Genetic algorithm. A comparative study has been made in the same computational environment. Figs. 9 and 10 show the comparison and results obtained using forest data structure algorithm with that of GA. It can be seen that the former performed better for all the three cases. In addition, we have also tested our model with different problem sizes as shown in Table 12.

It can be seen from the comparative study that for small data sets GA takes almost equal computational times as that of forest data structure algorithm. But, as the problem size increases, the forest data structure algorithm shows its outperformance over GA. In practical scenarios, one needs to work with large number of nodes hence the problem size increases to a very large extent. Therefore, the use of forest data structure algorithm will be appropriate in this case and hence uses of this algorithm in our model is justified.

Table 11
Different policies and their corresponding outcomes.

Problem size ($I^*J^*K^*M$)	Policy	Total cost (in $\times 10^5$ \$)	Total carbon emitted (in ton)	Optimal value of binary decision variables
5*10*20*5	Firm with strict carbon caps	229.96	689.31	$W_i=[11101]$ $Y_j=[1110010111]$ $Z_m=[01011]$
	Firm with carbon tax	221.57	731.23	$W_i=[10111]$ $Y_j=[1110010101]$ $Z_m=[01001]$
	Firm with carbon cap-and-trade	220.32	738.75	$W_i=[11001]$ $Y_j=[1111010101]$ $Z_m=[01110]$

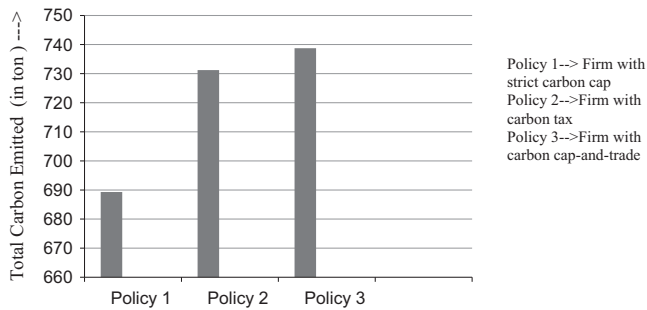


Fig. 4. Comparison of total carbon emitted under three proposed policies.

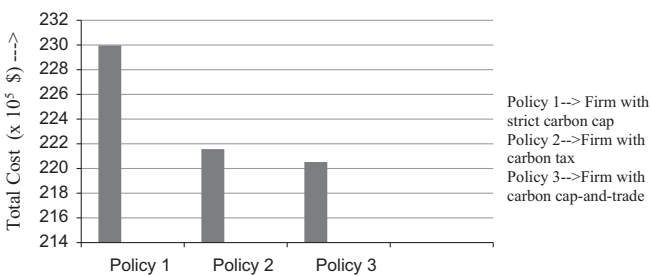


Fig. 5. Comparison of total costs under three proposed policies.

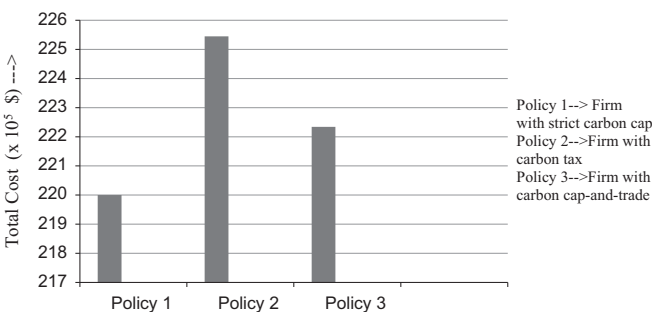


Fig. 6. Comparison of total costs under three proposed policies with different parameters as $CO_2^{CAP\ FORWARD} = 700$ t, $CO_2^{CAP\ REVERSE} = 200$ t, Carbon credit bought (b^+) = 100 t, Carbon credit sold (s^-) = 35 t.

7. Conclusions and scope of future research

With increasing pressure due to various rules and regulations on low carbon development, it has become important as well as challenging for firms worldwide to incorporate carbon footprint management into their business decisions. In this paper we presented a series of models considering different policies to show how carbon footprint considerations can be incorporated into quantitative operational models. We showed how decision makers can arrive at an optimal network configuration which minimizes

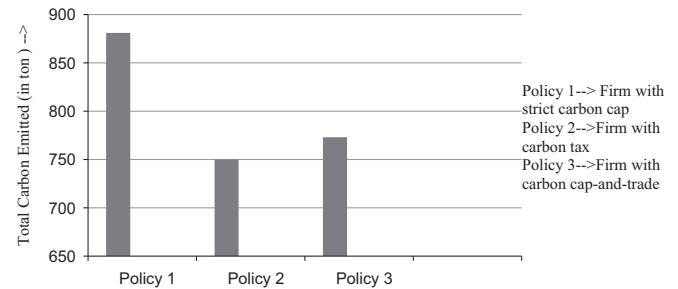


Fig. 7. Comparison of total carbon emitted under three proposed policies with different parameters as $CO_2^{CAP\ FORWARD} = 700$ t, $CO_2^{CAP\ REVERSE} = 200$ t, Carbon credit bought (b^+) = 100 t, Carbon credit sold (s^-) = 35 t.

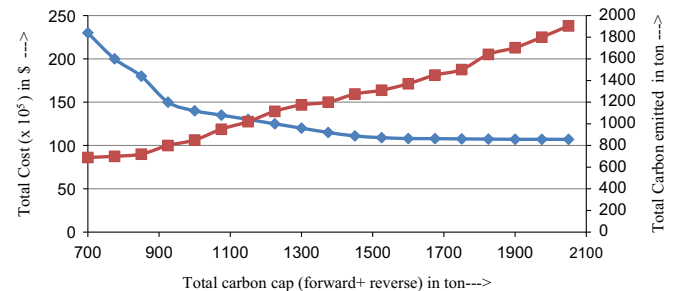


Fig. 8. Variation of total cost and total carbon emitted with carbon cap for a firm with strict carbon cap.

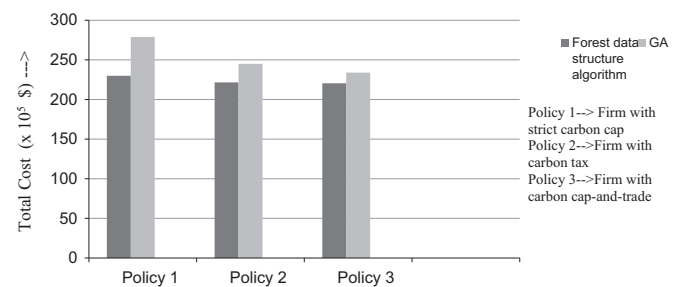


Fig. 9. Comparison of results obtained from forest data structure algorithm and GA for total cost.

both total cost and carbon footprint of the whole supply chain. This research showed how a firm could benefit by choosing proper combination among different parameters and different policies. From the model we arrived at one interesting observation that it is possible to significantly reduce carbon emissions without significantly increasing the total cost.

There are numerous avenues for future research. We will highlight a few of them and by doing so, it will simply emphasize the large number of operational problems that could be integrated with carbon footprint considerations. In our model, we have considered all the parameters to be deterministic but in many practical

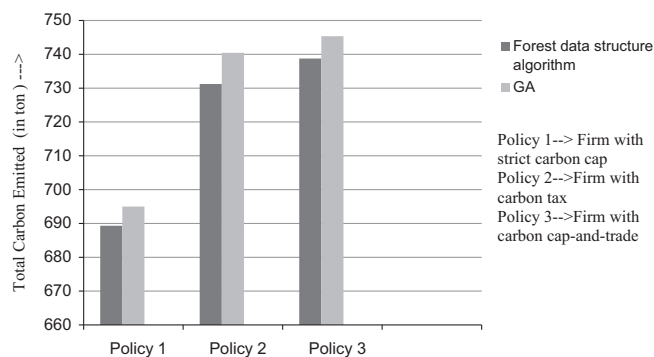


Fig. 10. Comparison of results obtained from forest data structure algorithm and GA for total carbon emitted.

Table 12

Comparison of computational times of two algorithms for different problem sizes.

Problem size ($F^*J^*K^*M$)	Computational time for forest data structure algorithm (in s)	Computational time for GA (in s)
3*5*7*3	0.45	0.44
3*10*7*3	0.49	0.50
5*10*10*3	0.67	0.78
5*10*20*5	0.72	1.56
7*20*30*5	0.87	1.98
7*20*30*7	0.91	2.34
10*39*50*9	1.03	2.89

scenarios customer demand, rate of return of products, etc. could vary a lot as time progresses. Therefore, we need to consider stochastic environment, perform scenario analysis and observe how the optimal network configuration changes. Also, for an optimal network configuration predicted by our model, one can apply sensitivity analysis to verify what are the ranges of different parameters for which the optimal network configuration remains optimal. This will show the robustness of the predicted network configuration. In this paper we have dealt with single product but one may introduce multi-products, concept of joint-ordering also. One may also consider multiple-firms and integrate the concept of collaboration with this model. Nowadays, many firms are having their branches situated in different countries across the world. Also, policies related to carbon footprint vary from one country to another. Therefore, it will be of interest to compare different policies and take operational decisions (for a particular branch of the firm situated in a particular country) accordingly so that the firm can generate maximum revenue and minimize total carbon emissions throughout its whole supply chain.

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